Foundation of Machine Learning 2025

Forecasting US Stock Closing Dynamics

This project investigates the dynamics of the Nasdaq Closing Cross auction using high-frequency order book and auction data from the Optiver Trading at the Close dataset.

Nael Arnoux Marta Shkreli Wei Luo - Machine Learning on High-Frequency Auction Data

Motivation & Context

Why the Closing Auction?

The Nasdaq Closing Cross determines the official closing price

- High volume, high volatility, information-dense
- Critical for:
  - Index calculation
  - Portfolio valuation
  - Algorithmic execution

Key Challenge

- Predicting short-term price movements in a noisy, fast-evolving market phase

Dataset Overview

Data Source

Tom Forbes, John Macgillivray, Matteo Pietrobon, Sohier Dane, and Maggie Demkin. Optiver - Trading at the Close. https://kaggle.com/competitions/optiver-trading-at-the-close, 2023. Kaggle.

Structure

- High-frequency snapshots per stock & trading day
- Final target: short-horizon price movement

Key Variables

- Prices: bid, ask, WAP, reference price
- Auction info: imbalance size & direction, matched size
- Time structure: seconds_in_bucket, time_id

Dataset Overview

stock_id	date_id	seconds_in_	imbalance_s	imbalance_b	reference_pr	matched_siz	far_price	near_price	bid_price	bid_size	ask_price	ask_size	wap	target	time_id	row_id
0	0	0	3180602.69	1	0.999812	13380276.64			0.999812	60651.5	1.000026	8493.03	1.0	-3.029704	0	0_0_0
1	0	0	166603.91	-1	0.999896	1642214.25			0.999896	3233.04	1.00066	20605.09	1.0	-5.519986	0	0_0_1
2	0	0	302879.87	-1	0.999561	1819368.03			0.999403	37956.0	1.000298	18995.0	1.0	-8.38995	0	0_0_2
3	0	0	11917682.27	-1	1.000171	18389745.62			0.999999	2324.9	1.000214	479032.4	1.0	-4.0102005	0	0_0_3
4	0	0	447549.96	-1	0.999532	17860614.95			0.999394	16485.54	1.000016	434.1	1.0	-7.3498487	0	0_0_4
5	0	0	0.0	0	1.000635	13552875.92			0.999779	1962.72	1.000635	5647.65	1.0	6.7794323	0	0_0_5

- The dataset contains high-frequency order book and auction data for U.S. equities during the closing auction period.
- Each row represents the state of the order book at a specific second before the market close.
- Data spans multiple stocks (**stock_id**) and trading days (**date_id**).

Feature Engineering (Market Microstructure)

Core Engineered Features

- **Directional Imbalance Pressure:** Combines size + direction of auction imbalance
- **Bid–Ask Spread:** Proxy for liquidity and uncertainty
- **Auction Price Divergence:** Difference between far and near prices

Dynamic Features

- First differences (momentum):
  - Imbalance size diff
  - Matched size diff
- Lagged variables (for LightGBM)

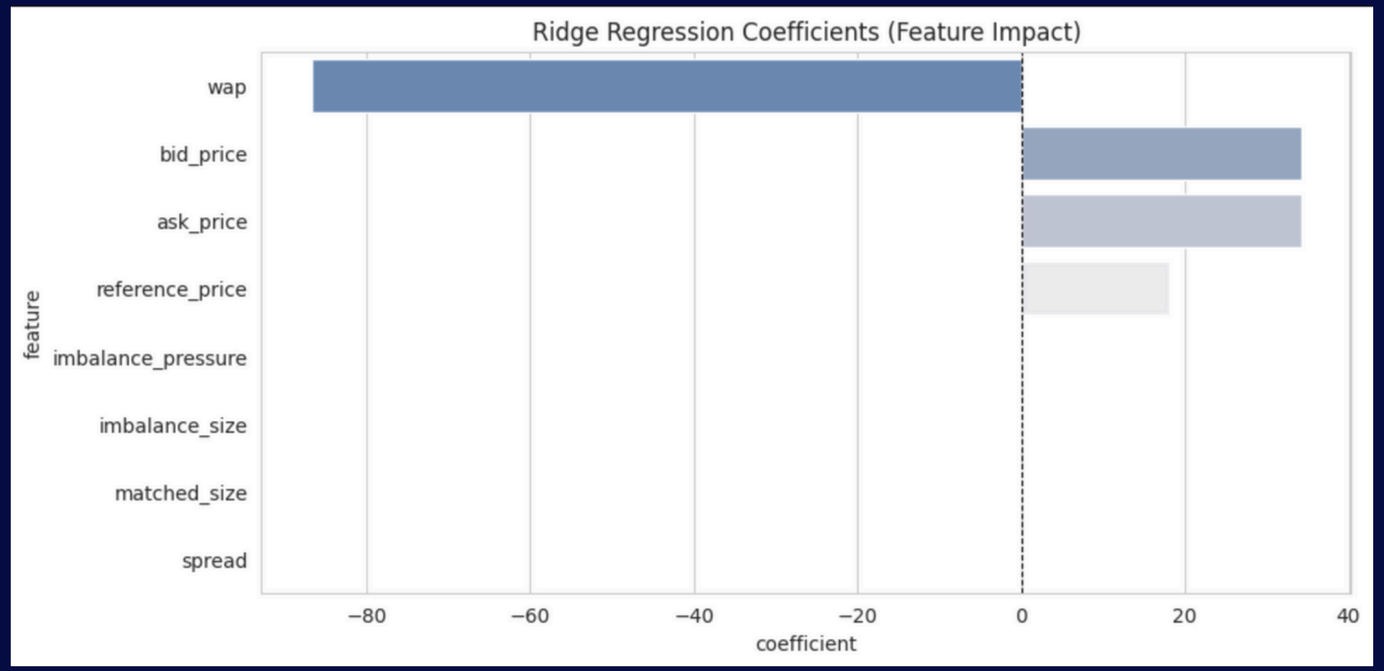
Ridge Regression (Baseline)

Why Ridge?

- Handles multicollinearity
- Stable linear baseline
- High interpretability

Features Used

- Prices: bid, ask, WAP, reference
- Volumes: matched size, imbalance size
- Engineered: spread, imbalance pressure



Metric	Value
MAE	6.310143
RMSE	9.286936

Random Forest Model

Why Random Forest?

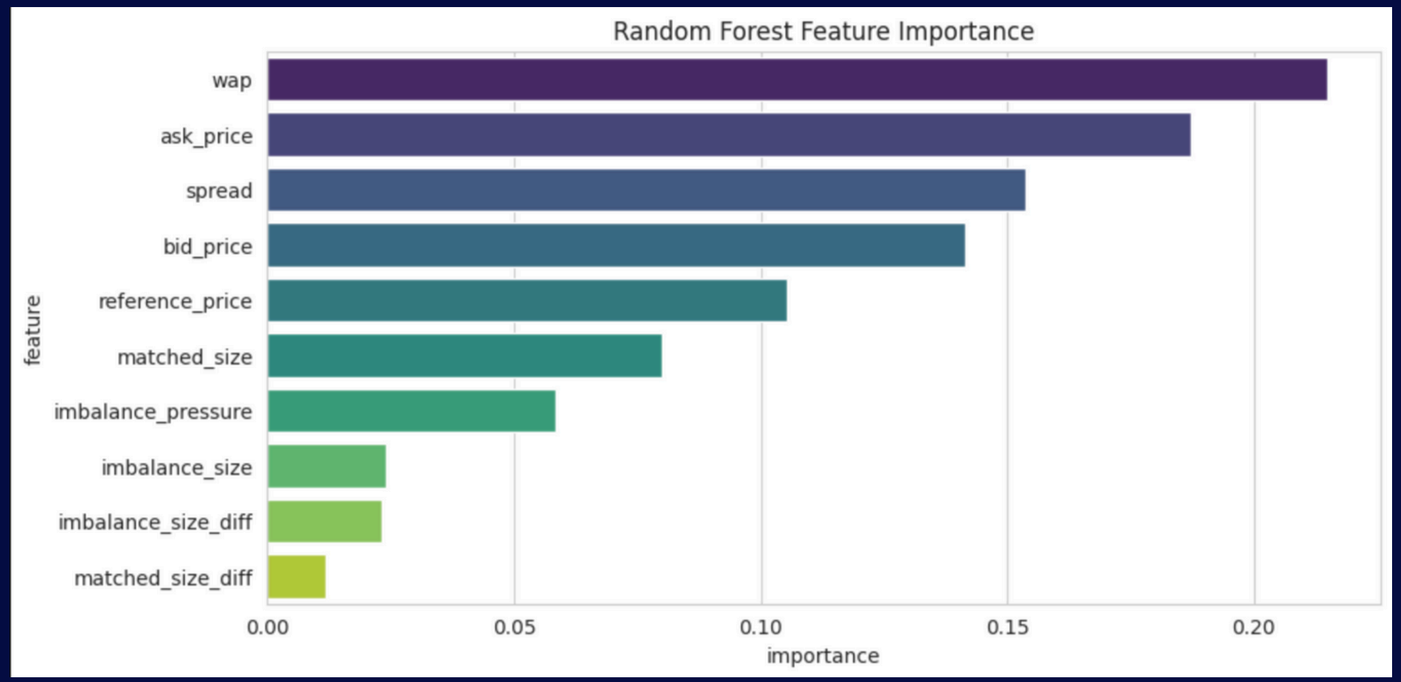
- Captures nonlinear relationships
- Handles feature interactions automatically

Additional Features

- Momentum features (differences)
- Same microstructure signals as Ridge

Observation

- Slightly worse than Ridge
- Nonlinear effects add limited predictive power



Metric	Value
MAE	6.448341
RMSE	9.848583

LightGBM Model (Best Performer)

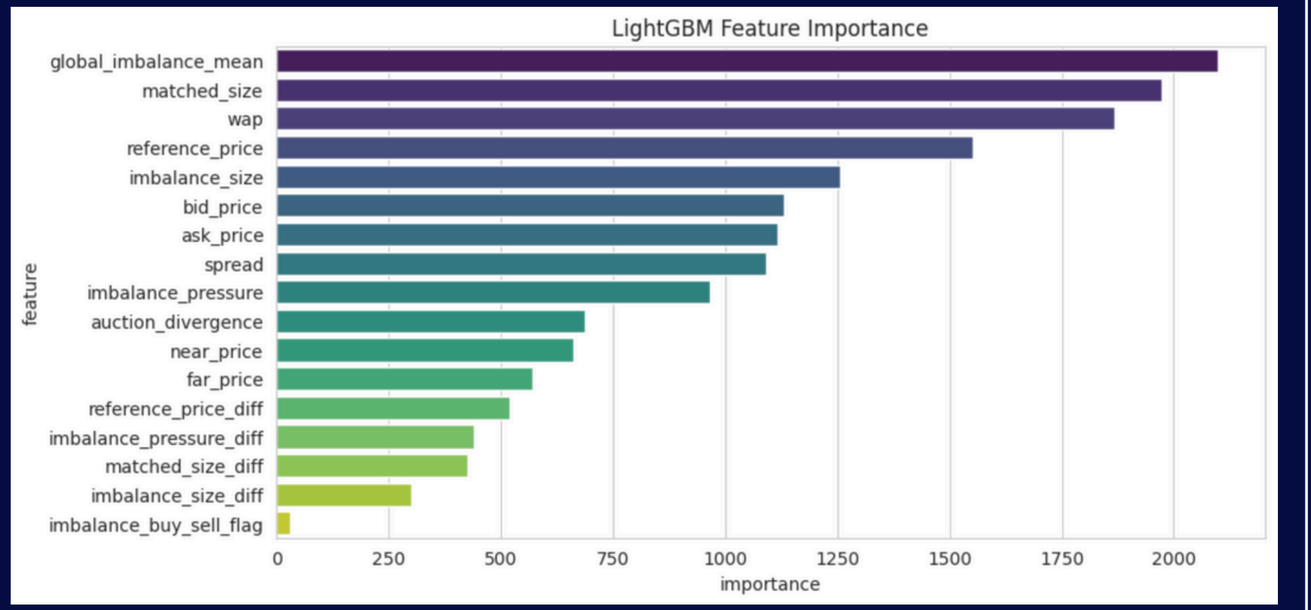
Why LightGBM?

- Gradient boosting → strong nonlinear modeling
- Efficient on large, high-frequency datasets
- Robust to extreme values (MAE loss)

Key Enhancements

- Lag-based temporal features
- Auction divergence
- Global imbalance mean (cross-sectional signal)
- Early stopping to prevent overfitting

~16.6% improvement over Ridge



Metric	Value
MAE	5.260126
RMSE	7.630974

Results & Interpretation

What drove the performance?

- **Price is King:** Bid_Price, Ask_Price, and WAP are the top predictors across all models. The market state at t is the best predictor of t+1.
- **The "Alpha" comes from Microstructure:**
  - Imbalance Pressure and Spread (our engineered features) ranked high in LightGBM
  - These features help the model predict deviations from the trend (volatility)

Performance Leaderboard (MAE)

- #1 - LightGBM: 5.26
- #2 - Ridge Regression: 6.31
- #3 - Random Forest: 6.45

Key Takeaway

- Linear models are good for the "trend," but Boosting (LightGBM) is required to minimize error during volatile auction phases.

Conclusion